

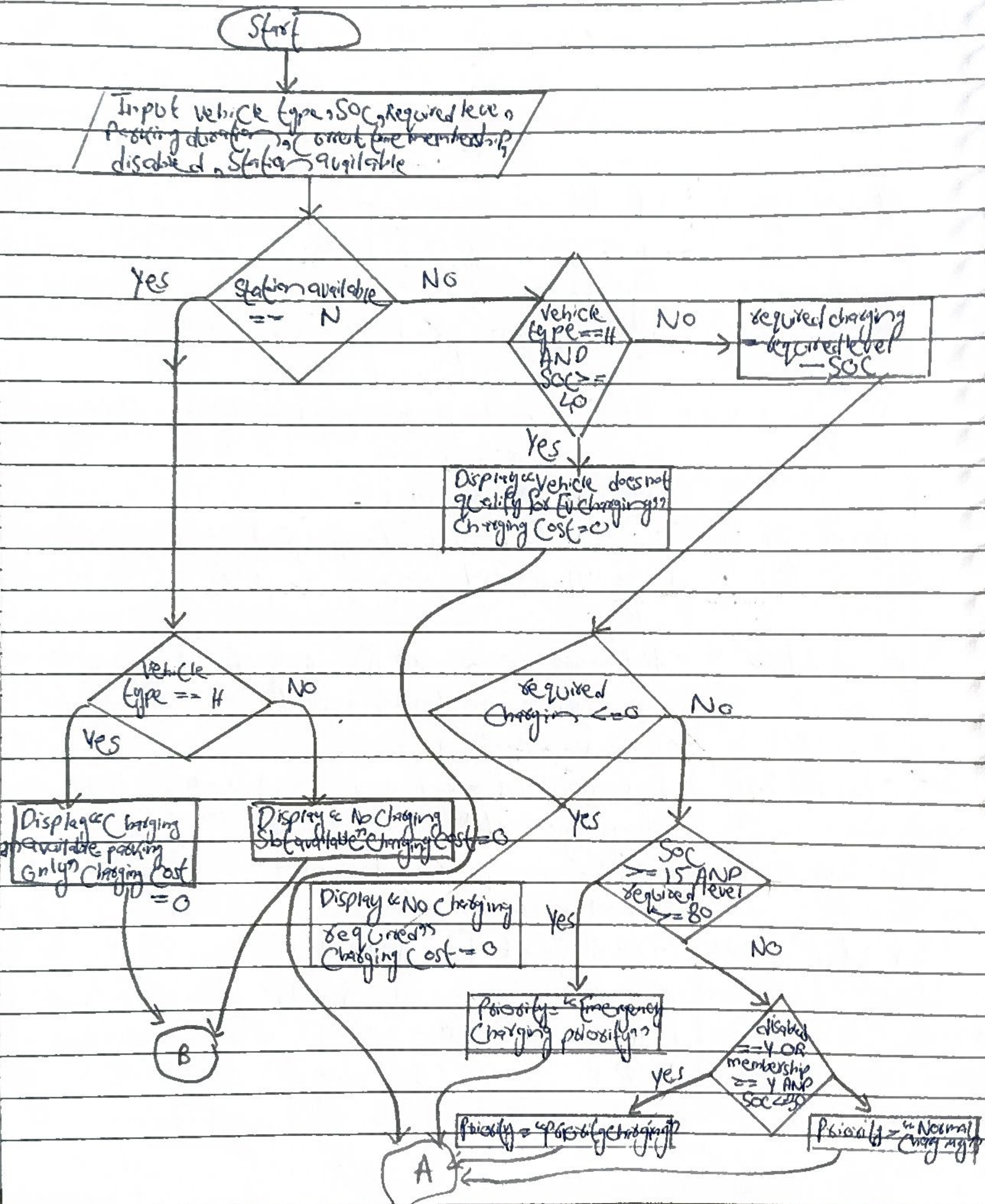
Question 6

PAC

Given Input	Output	Processing Required
Vehicle type (E/H), SOC(A) required level (%)	Charging priority (peak/ off)	Check station availability and vehicle type
parking duration (hours) Current time (24hrs)	Peak status, Charging cost, discount applied, final payable amount, warning	eligibility required $\geq$ required level - SOC
membership (Y/N), Disabled (Y/N), Station available (Y/N)	/ Status messages	Assign priority via nested conditions
		Determine peak/off peak rate and membership discount
		Charging Cost = required Charging $\times$ rate
		Determine parking base by duration, parking cost by disabled/membership status
		final payable = Charging Cost + parking Cost

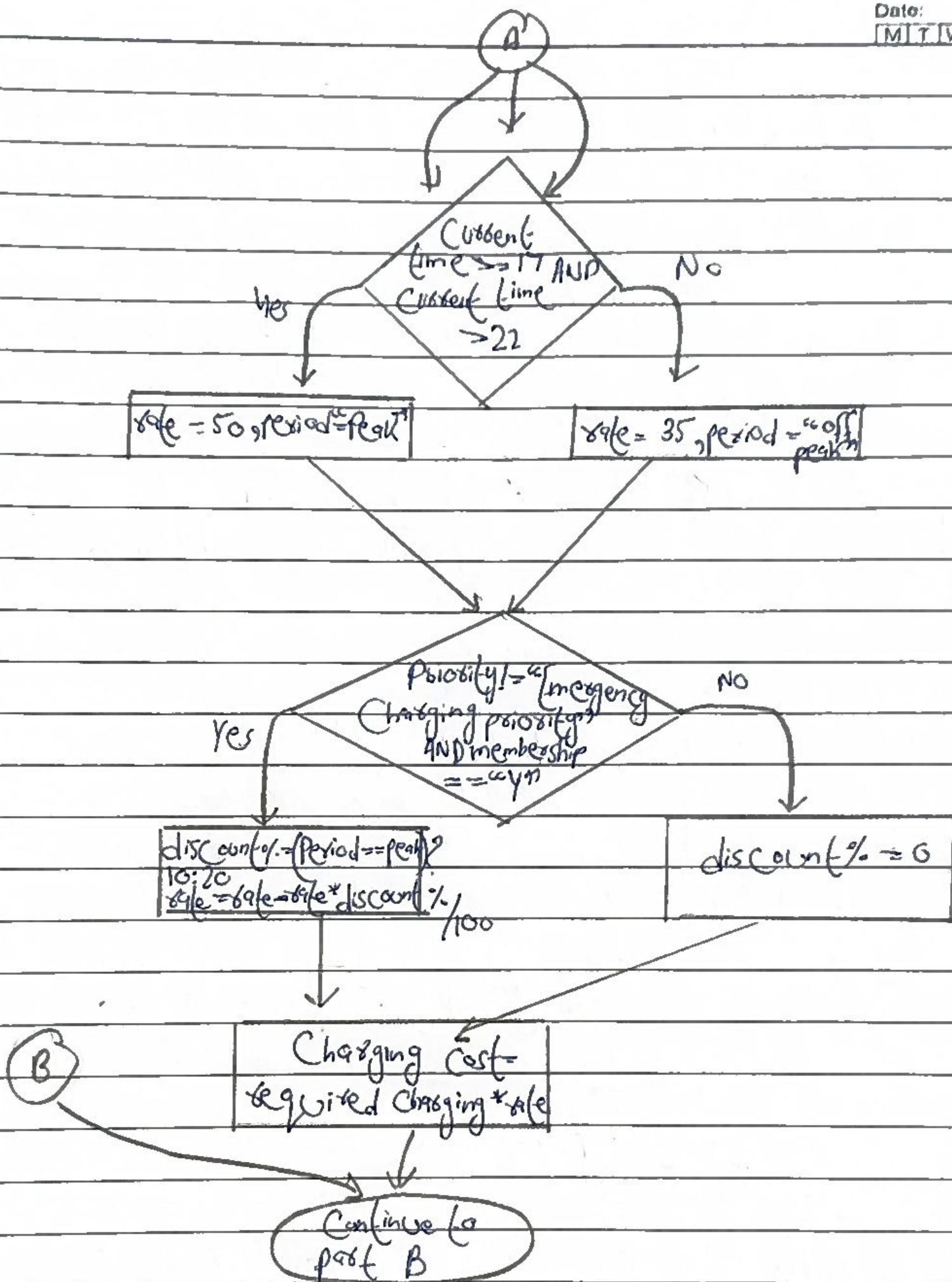
IPO

Input	Process	Output
1) Vehicle type	1) If station unavailable $\rightarrow$ display appropriate message (hybrid vs non hybrid), Skip charging	1) Vehicle type, SOC, required %
2) SOC	1) Hybrid qualifies only if $SOC < 40\%$ else "does not qualify"	1) Charging priority
3) Required level	1) Required charging $\neq$ required level - SOC; if $\leq 0 \rightarrow$ "No charging required"	1) Peak/Off Peak status
4) parking duration	1) Nested if $\&$ assign emergency/priority/Normal charging category	1) Charging cost, parking cost, discount
5) Current time	1) Determine peak or off peak rate; apply membership discount except for emergency	1) Final payable amount
6) membership	1) Charging Cost = required charging $\times$ rate	1) Warning status message
7) disabled	1) Determine parking base by duration	
8) Station available	1) Parking Cost: free if disabled; 20% off if member, else full rate 2) If duration > 8 hrs $\rightarrow$ long stay warning	

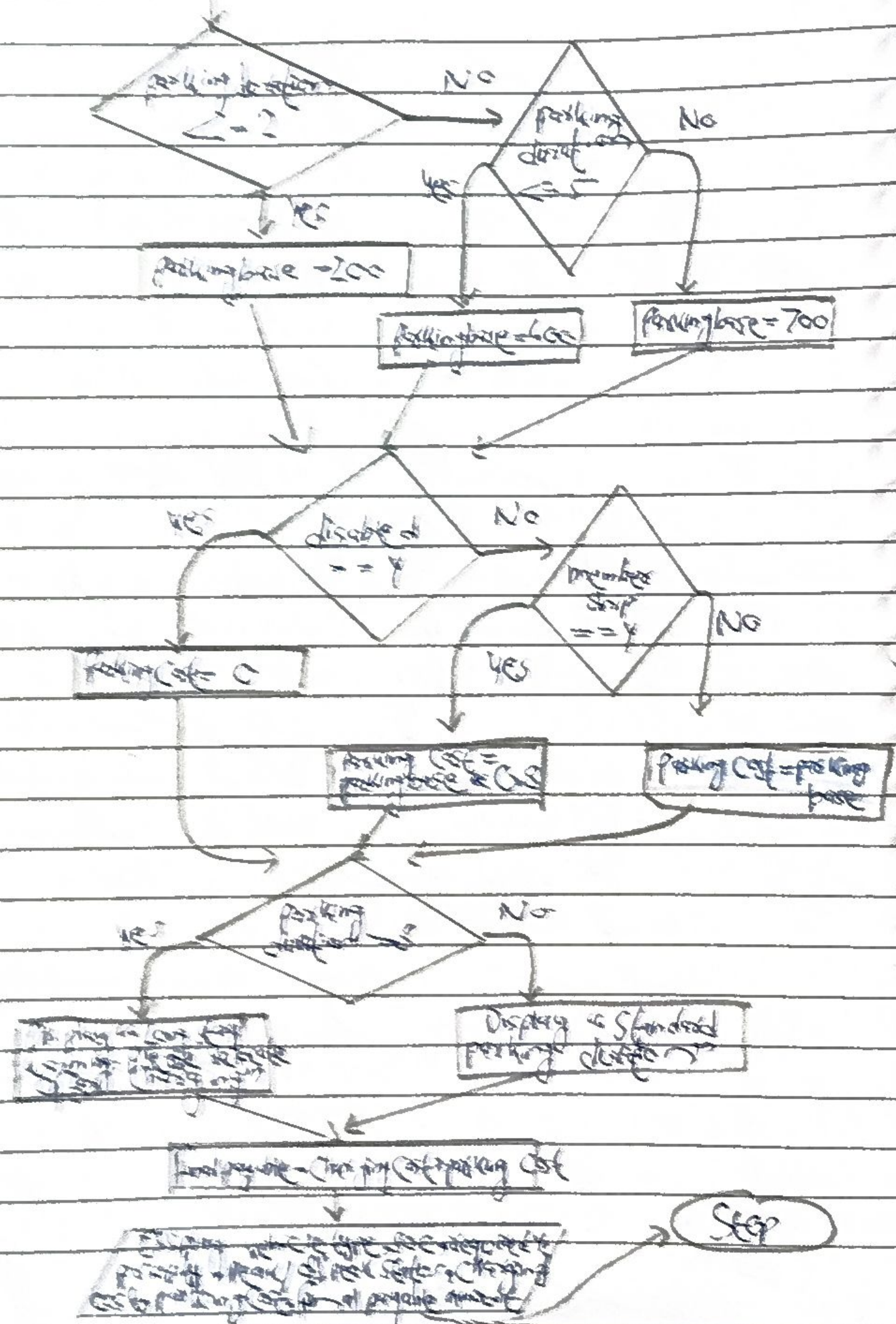


Date:

M T W T F S S



Example 3



Begin

Input vehicle type (ETH), SOC, required level, parking duration, current time, membership, disabled, station available

charging cost = 0
 priority = "N/A"
 period = "N/A"
 discount percent = 0

If
 Station available = "N" THEN
 If vehicle type = "H" THEN
 Display "Charging unavailable parking only"

Else
 Display "No charging slot available"

End If

While Display "Vehicle does not qualify for EV charging"

Else

Required = required level - SOC

If required charging ≤ 0 THEN

Display "No charging required"

Else

priority = "Emergency charging priority"

Else If disabled = "Y" OR (membership = "Y" AND SOC ≤ 30)

priority = "priority charging"

Else

priority = "normal charging"

End If

IF Current time ≥ 7 AND Current time ≤ 22

rate = 50

period = "Peak"

Else

rate = 35

period = "Off peak"

END IF

IF priority $\neq$ "Emergency charging priority" AND membership $\neq$ "y"

IF period $\neq$ "Peak"

discount percent = 10

Else

discount percent = 20

END IF

rate $\rightarrow$ rate - rate * (discount percent / 100)

END IF

END IF

IF parking duration ≤ 2

parking base = 200

Else if parking duration ≤ 5

parking base = 400

Else

parking base = 700

END IF

IF disabled = "y"

parking cost = 0

Else IF membership $\neq$ "y"

parking cost = parking base * 0.8

Else

parking Cost = parking base

End If

If parking duration > 8

Display "Long stay warning:- Please relocate your vehicle after parking"

Else

Display "Standard parking duration"

End If

final payable = charging cost + parking cost

Display "Vehicle type", Vehicle type

Display "Battery level (SOC)", SOC, "%"

Display "Charging priority", priority

Display "Period", Period

Display "Charging Cost Rs.", Charging Cost

Display "Parking Cost Rs.", parking Cost

Display "Required charging level", Required level, "%"

Display "Discount applied", discount percent, "%"

Display "Final payable amount", Rs final payable

End